

Decision-time forward pass: provenance-explicit inputs

Privileged offline / control
sources

Logged evidence

RGB, pose, calibration
semi-dense or fused
geometry
frozen EVL features /
support
observed target source
(v1)

Model state and target

s_t^{cf0} , admitted descriptor
 φ_e
selected-view history
remaining budget; source
named

Privileged scene assets

mesh, GT boxes, target
crop
identity and evaluation
support
GT descriptor source (v0)

v1: observed

v0: GT control

Candidate rows

finite poses $\mathcal{Q}_t$
protocol-legal row
features
proposal provenance
explicit

Q_H candidate scorer

f_θ on $(s_t, e, q_{t,i}, h)$
raw value and feasibility
logit
per materialized row

Candidate renders

hard-valid candidate
depth / points
target endpoint error

Hard action support

$m_{t,i}^{\text{act}}$
gates selection; feasibility
target
never changes raw Q_H

Predictions and action

raw $Q_{h,\theta,e,i}^{\text{cond}}$ and
feasibility
masked selection
downstream

Oracle targets

one-step target RRI
bounded returns and
upper bounds

Reason codes explain rejected
rows; they are neither
embeddings nor low rewards.

feasibility target

Training / evaluation only

prediction-target
comparison
value / feasibility loss,
metrics
named ablations

value supervision

Q -label support

$m_{t,i}^{Q,h} \leq m_{t,i}^{\text{act}}$
finite value-target rows
gates value loss and
successor backup